In this Learning Module you will learn about **prime numbers**, **prime factors**, and **prime factorization**. In order to do this, you will first need to understand what **prime numbers** and **factors** are. Read the definition below.

prime number

A **prime number** is a number **greater than 1** having only **two factors,** 1 and itself!

Go back and reread the definition above, using your marker to circle the words in the definition that tell you what a **prime number** is.

One way to remember this definition is to think of the word **prime** as shown below.

prIME

The "i" in the word becomes the factor 1 and the "me" at the end of the word stands for the second factor, the number itself.

3 is **prime** because it has only 1 and 3 as factors.

3 is prIME

Now that you understand what a **prime number** is, you will also need to understand what **factors** are. Read the definition below.

Factor

In general, a **factor** is a number that is multiplied by another number in a multiplication situation resulting in a product.

For example, in the equation 3 x 8 = 24, the numbers **3** and **8** are **factors**.

Because 3 and 8 are **factors** of 24, this means that the number 24 can also be evenly divided by both 3 and 8.

Go back and reread the definition above, using your marker to circle the words in the definition that tell you what a **factor** is.

In the problems below, the **factors** have been circled. Use your marker to trace the circles around the **factors**.

$$\begin{array}{r} 5 \\ \times\ 8 \\ \hline 40 \end{array} \qquad 7 \times 6 = 42 \qquad \begin{array}{r} 4 \\ \times\ 9 \\ \hline 36 \end{array}$$

Use your marker to write the word **factor** in each box below.

factor

8 x 7 = 56

factor

factor

9 x 6 = 54

factor

factor

→12

x 6 ←

72

factor

factor

factor

2 x 3 x 4 = 24

factor

Factors

Follow the directions.

One way to identify the first dozen **prime numbers** is to use a 1-40 Chart.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40

1) Cross out the number **1**, as **prime numbers** must be **greater than 1**.

2) Circle 2 because the only factors of 2 are 1 and 2. **Cross out** all of the other **even numbers**, because 2 is a factor for every even number.

3) Circle 3 because the only factors of 3 are 1 and 3. **Cross out** all of the other **multiples of 3** because 3 is a factor of its multiples.

4) Circle 5 because the only factors of 5 are 1 and 5. **Cross out** all of the other **multiples of 5** because 5 is a factor of its multiples.

5) Circle 7 because the only factors of 7 are 1 and 7. **Cross out** all of the other **multiples of 7** because 7 is a factor of its multiples.

Continue circling the **prime numbers** in the chart.

x	2	3	x	5	x	7	x	x	x
11	x	13	x	x	x	17	x	19	x
x	x	23	x	x	x	x	x	29	x
31	x	x	x	x	x	37	x	x	x

6) **Circle 11** because the only factors of 11 are 1 and 11.

7) **Circle 13** because the only factors of 13 are 1 and 13.

8) **Circle 17** because the only factors of 17 are 1 and 17.

9) **Circle 19** because the only factors of 19 are 1 and 19.

10) **Circle 23** because the only factors of 23 are 1 and 23.

11) **Circle 29** because the only factors of 29 are 1 and 29.

12) **Circle 31** because the only factors of 31 are 1 and 31.

13) **Circle 37** because the only factors of 37 are 1 and 37.

The circled numbers in the chart represent the first 12 **prime numbers!**

On the previous pages you found the first 12 **prime numbers**. These are the most commonly used **prime numbers**, so it is a good idea to simply memorize them. A good way to memorize something is to write and say the information to be learned.

Below is the list of the first twelve **prime numbers**.

2 , 3 , 5 , 7 , 11 , 13 , 17 , 19 , 23 , 29 , 31 , 37

Copy the list on each of the lines below. Say the numbers quietly to yourself as you write them.

1) _____

2) _____

3) _____

4)

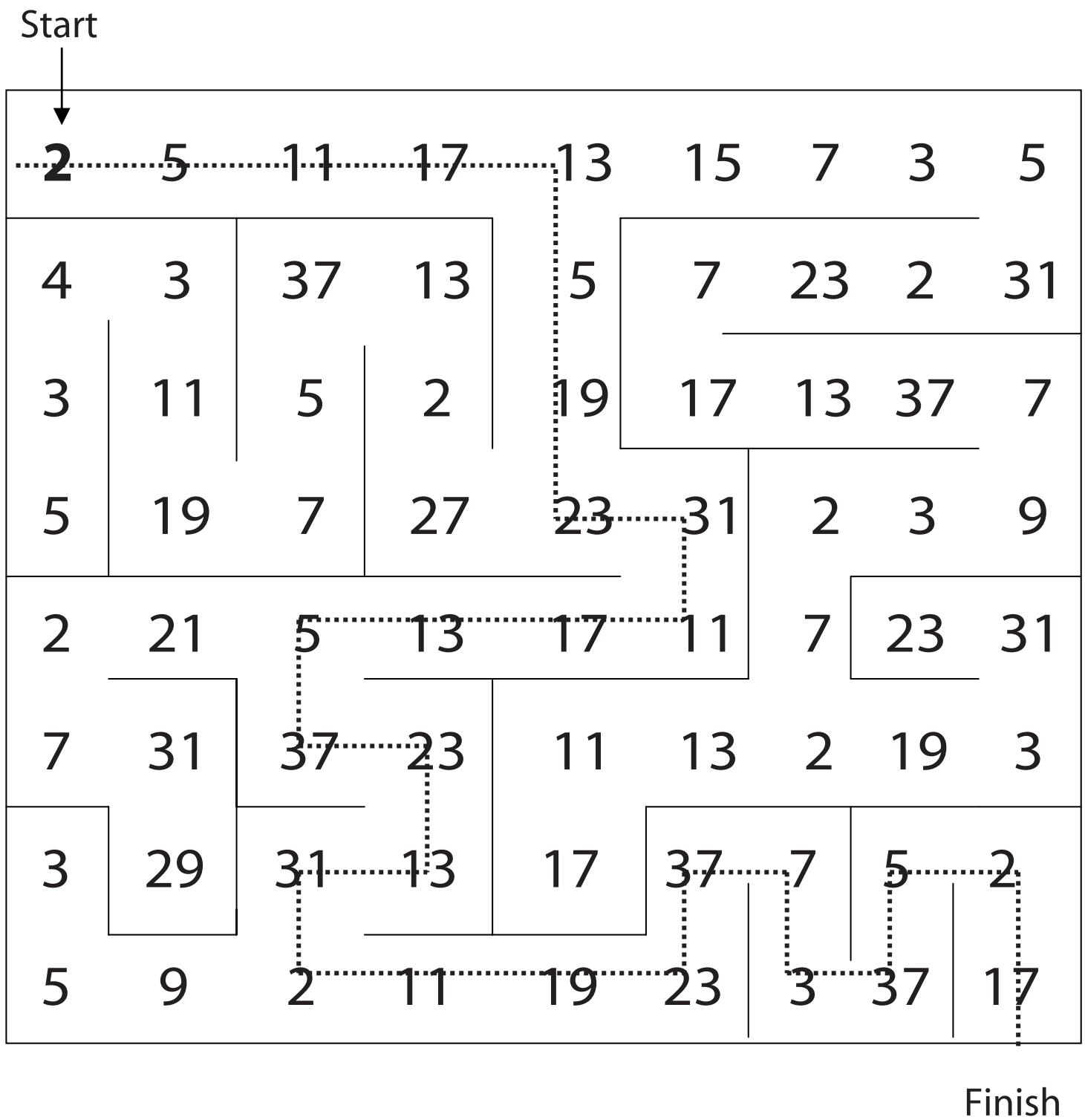
5)

Follow the path of only **prime numbers** to get through the maze.

Identifying Prime Numbers *Follow the directions.*

Follow the path of only **prime numbers** to get through the maze.

Trace over the path through the maze with your marker, saying the numbers to yourself as you go.



Now that you know what **prime numbers** are, you can use this knowledge to learn how to find the **prime factors** for whole numbers. Remember that a **factor** is a number that is multiplied by another number in a multiplication situation resulting in a product.

The list below shows all of the whole number multiplication equations that give a product of 12.

The two numbers being multiplied in each equation are called a **factor pair.**

$1 \times 12 = 12$

$2 \times 6 = 12$

$3 \times 4 = 12$

$4 \times 3 = 12$

$6 \times 2 = 12$

$12 \times 1 = 12$

The numbers being multiplied in these equations are 1, 2, 3, 4, 6, and 12.

The **factors** of 12 are 1, 2, 3, 4, 6, and 12.

Circle the **factors** of 12 above that are **prime numbers**.

The previous page listed the factors of 12, but not all of those **factors** were **prime numbers**.

To find the **prime factors** of a number, you can start with any of the **factor pairs** of that number except the pair containing 1 and the number itself. Remember, that a **prime number** is a number **greater than 1**.

~~1 x 12 = 12~~

2 x 6 = 12

3 x 4 = 12

4 x 3 = 12

6 x 2 = 12

~~12 x 1 = 12~~

Circle the equations below containing the **factor pairs** that you can use to begin finding the **prime factors** of 12

2 x 6 = 12

3 x 4 = 12

4 x 3 = 12

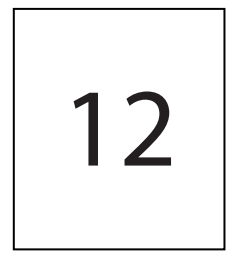
6 x 2 = 12

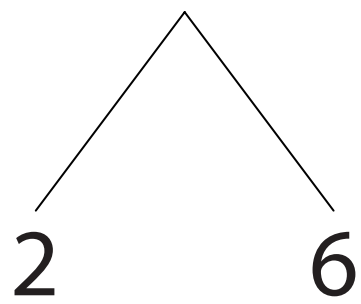
An easy way to find the **prime factors** of a number is to use a **factor tree**. A **factor tree** begins with the number to be factored.

12

Then you add two "branches" below the number, using any of the **factor pairs** except those containing 1 and the number.

1) Trace the number 12. →

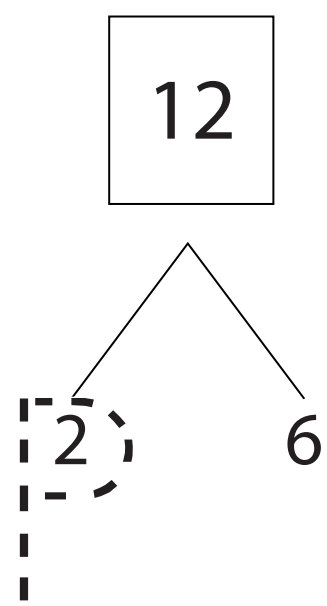




← 2) Trace the branches leading to the factor pair being used.

3) Trace the 2 and the 6. →

Check to see if either of the "branch" numbers is **prime**.
Put a "P" around the numbers that are **prime**.



4) Trace the letter P P drawn around the 2 →

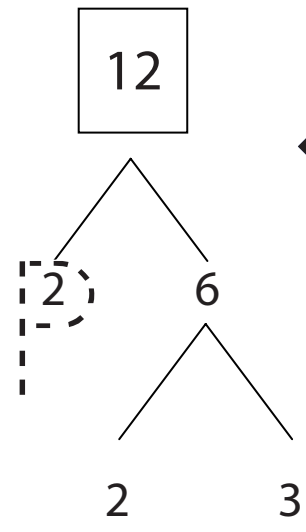
Prime Factors

Read the information and follow the instructions.

Draw 2 new "branches" under the 6 (because 6 is **not** a **prime number** and write the two **factors** of 6 greater than 1.

1) Trace the number 12. →

3) Trace the 2 and the 6. Trace the letter "P" drawn around the 2. →



← 2) Trace the "branches" leading to the factor pair being used.

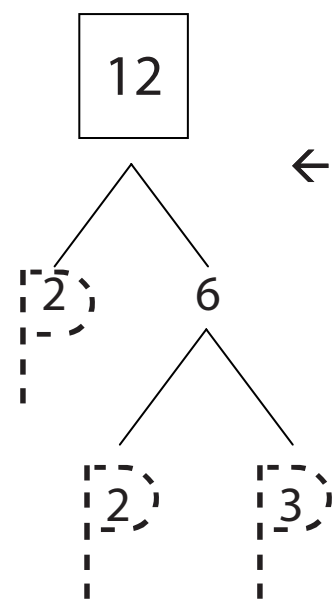
← 4) Trace the "branches" and the new factor pair.

Check to see if either of the "branch" numbers is **prime**.
Put a "P" around the numbers that are **prime**.

1) Trace the number 12. →

3) Trace the 2 and the 6. Trace the letter P drawn around the 2. →

5) Trace the letter P drawn around the 2 and the 3. →



← 2) Trace the branches leading to the factor pair being used.

← 4) Trace the branches and the new factor pair.

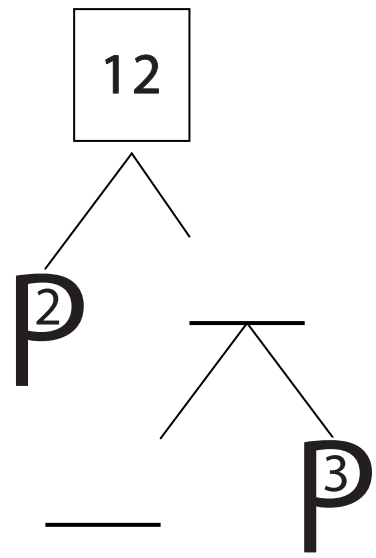
Write the numbers in the "P"s (least to greatest) as **factors** in an expression.

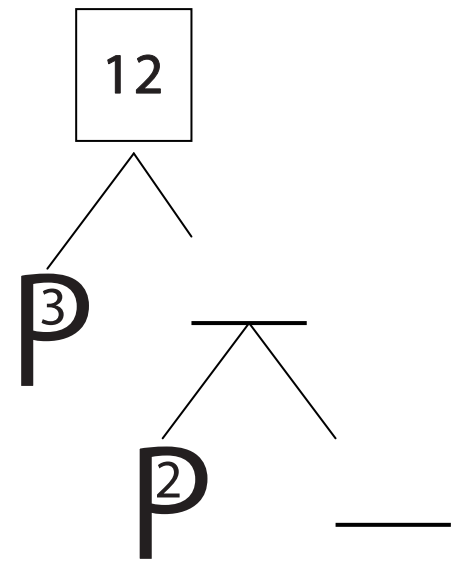
2 x 2 x 3

Prime Factors

Read the information and follow the instructions.

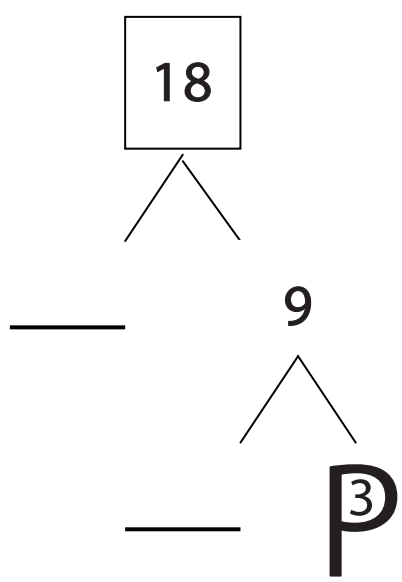
Complete the **factor trees** below for the number 12.

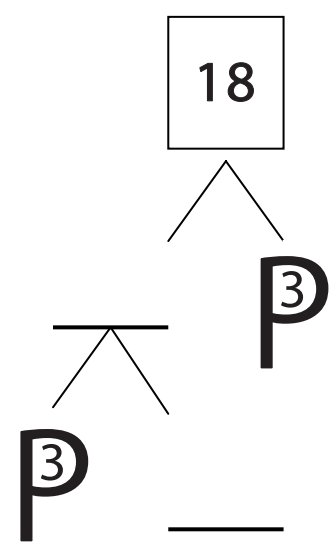




What are the **prime factors** of 12? _____

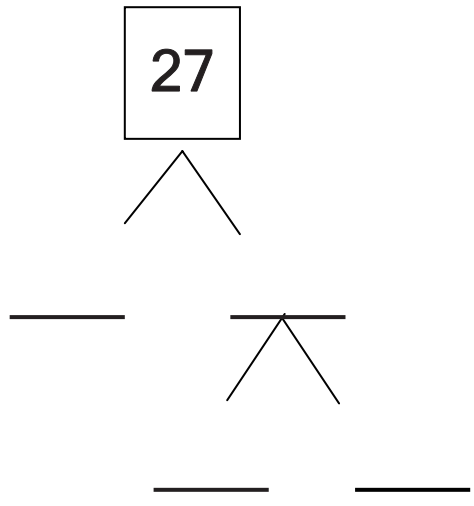
Complete the **factor trees** below for the number 18.





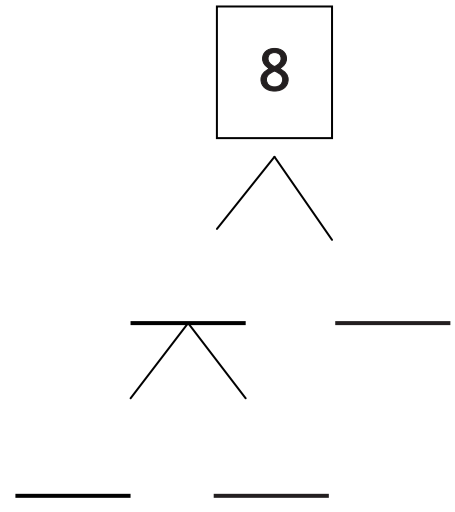
What are the **prime factors** of 18? _____

Complete the **factor trees** and list the **prime factors** for each number.



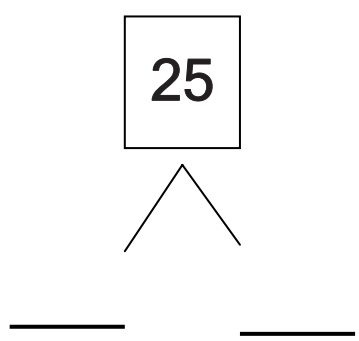
What are the **prime factors** of 27?

_____



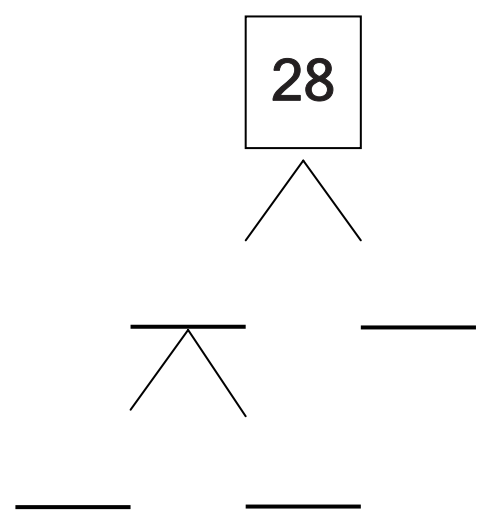
What are the **prime factors** of 8?

_____



What are the **prime factors** of 25?

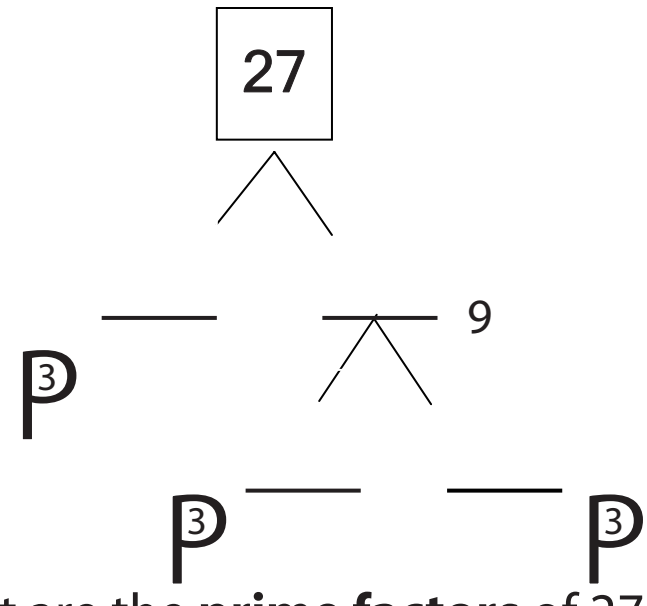
_____



What are the **prime factors** of 28?

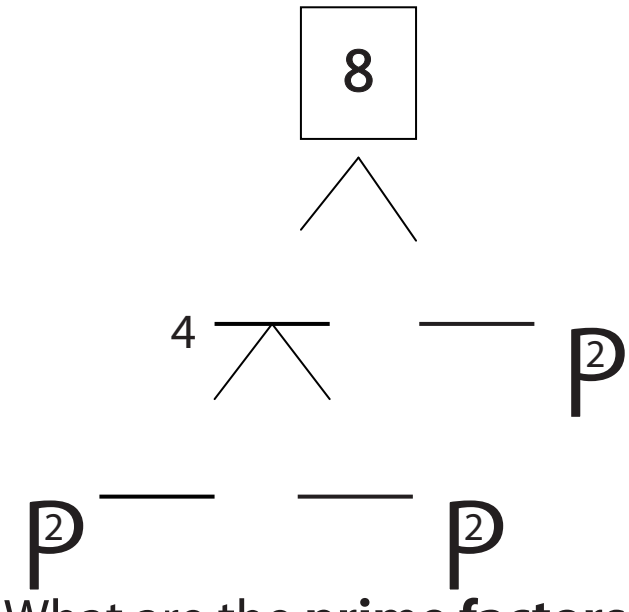
_____

Check your work. Go back to Q12 and try again if you made any mistakes.



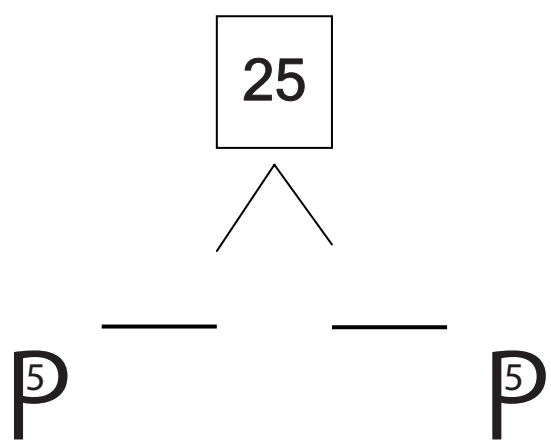
What are the **prime factors** of 27?

3 x 3 x 3



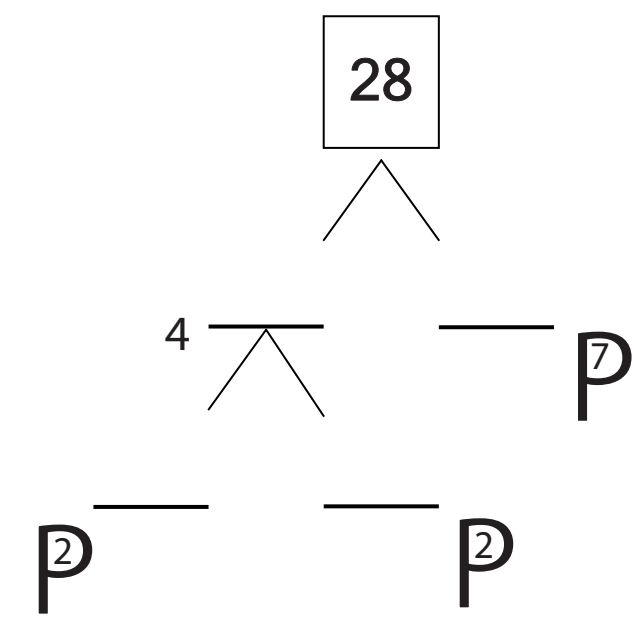
What are the **prime factors** of 8?

2 x 2 x 2



What are the **prime factors** of 25?

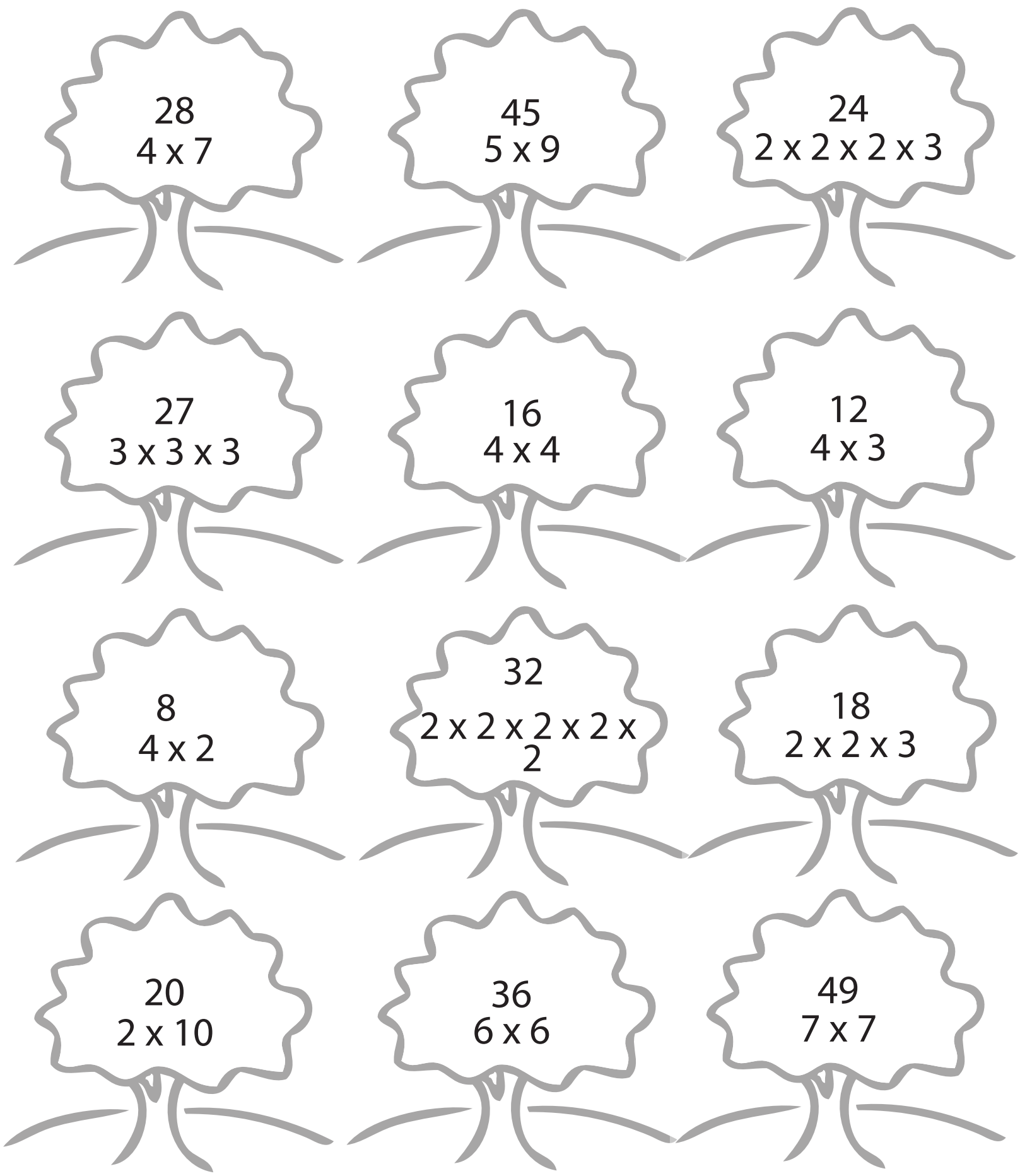
5 x 5



What are the **prime factors** of 28?

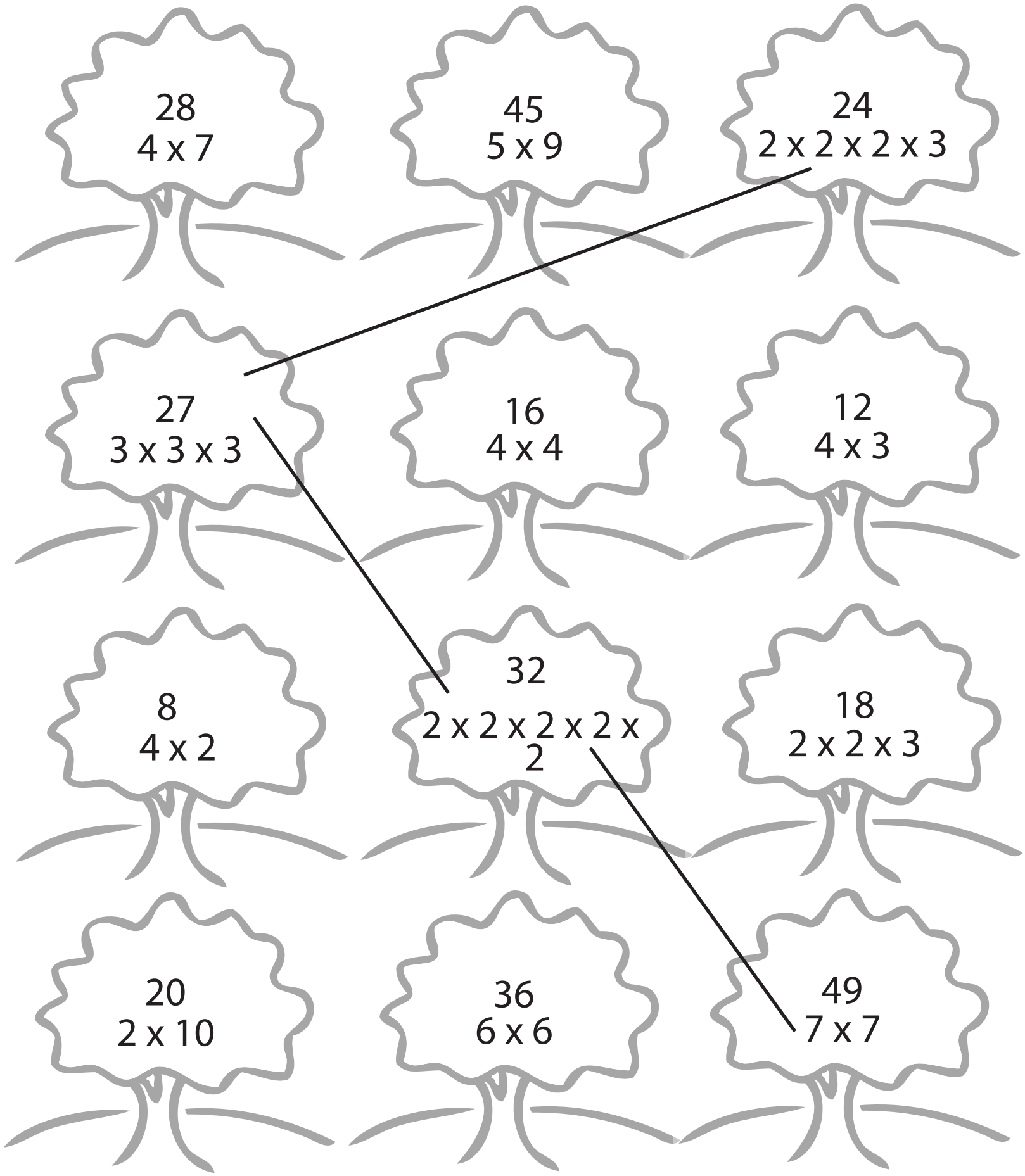
2 x 2 x 7

Make a path through the forest by connecting only the trees showing the correct prime factorization for each tree's number.



Prime Factors *Follow the directions.*

Did you find the correct path of **prime factorizations**? If not, go back to Q16 and try it again.



Sometimes you may have to answer multiple choice questions involving **prime numbers** or **prime factorizations**. The important thing to remember is that each answer choice is there for a reason. Two good strategies for choosing the correct answer are to **eliminate answer choices** and to **do your own work** first and find the answer choice that matches your work!

1) Which of the following is a list containing only prime numbers?

A 2 , 3 , 5 , 7 , 9 , 13 **9 is NOT prime, so A can be eliminated**

B 5 , 7 , 11 , 13 , 15 , 17 **15 is NOT prime, so B can be eliminated**

C 7 , 11 , 13 , 17 , 19 , 23 **All numbers in this list are prime**

D 2 , 5 , 11 , 13 , 19 , 21 **21 is NOT prime, so D can be eliminated**

Circle the correct answer choice for problem 1 then read and circle the correct answer for number 2 below.

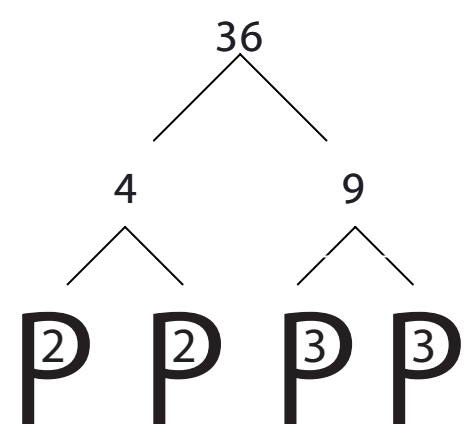
1) Which of the following is a list containing only prime numbers?

A 2 x 2 x 9

B 2 x 3 x 4

C 3 x 3 x 4

D 2 x 2 x 3 x 3



Use the strategies then circle the letter for the correct answer to each question.

1) Which of the following is a list containing only prime numbers?

A 1 , 2 , 3 , 5 , 7 , 11

B 2 , 3 , 7 , 9 , 13 , 19

C 5 , 7 , 13 , 17 , 23 , 27

D 3 , 7 , 13 , 17 , 19 , 23

2) Which of the following is NOT a prime number?

A 17

B 19

C 21

D 23

3) Which of the following shows 54 as the product of prime factors?

A 2 x 2 x 3 x 3

B 2 x 3 x 3 x 3

C 2 x 3 x 9

D 3 x 3 x 6

1) D 2) C 3) B

It's time to show how much you have learned about **prime numbers** and **prime factors**. Use the strategies then circle the letter for the correct answer to each question.

1) Which of the following is a list containing only prime numbers?

A 37, 31, 21, 19, 17, 13

B 31, 23, 19, 17, 13, 11

C 38, 37, 31, 23, 19, 17

D 37, 33, 31, 29, 23, 19

2) Which of the following is NOT a prime number?

A 39 **C** 31

B 37 **D** 29

3) Which of the following shows 72 as the product of prime factors?

A 2 x 2 x 2 x 9

B 2 x 3 x 3 x 3

C 2 x 2 x 3 x 6

D 2 x 2 x 2 x 3 x 3

Check your work. Did you answer all of the questions correctly? If not, go back to Q 20 and try again.

1) Which of the following is a list containing only prime numbers?

A 37, 31, 21, 19, 17, 13

B 31, 23, 19, 17, 13, 11

C 38, 37, 31, 23, 19, 17

D 37, 33, 31, 29, 23, 19

2) Which of the following is NOT a prime number?

A 39

B 37

C 31

D 29

3) Which of the following shows 72 as the product of prime factors?

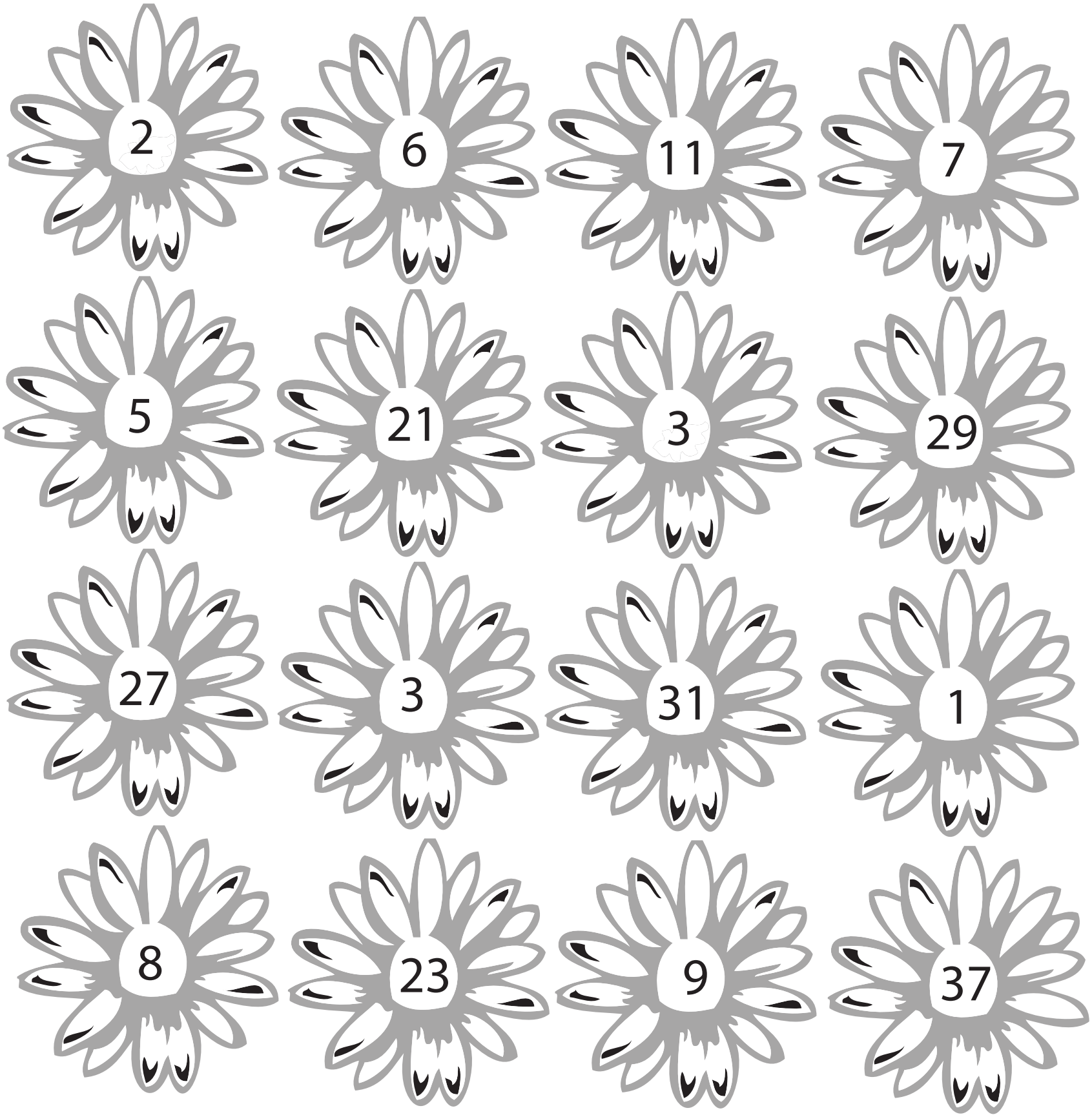
A 2 x 2 x 2 x 9

B 2 x 3 x 3 x 3

C 2 x 2 x 3 x 6

D 2 x 2 x 2 x 3 x 3

The bee can only collect pollen from "**prime**" flowers. Put an X on every flower from which the bee could collect pollen.



After checking your work, **erase** any marks and go on to the next page

2 11 7 5 3 29 3 31 23 37

Prime Numbers *Follow the directions.*

The number 30 when written as the product of **prime factors** is

$$2 \times 3 \times 5$$

The **prime factorization** of 30 doesn't have any **factors** that repeat, so it must always be written this same way.

The number 32 when written as the product of **prime factors** is

$$2 \times 2 \times 2 \times 2 \times 2$$

The **prime factorization** of 32 repeats the same **factor** 5 times.

Trace the 2 s in the prime factorization of 32 shown above.
Count how many 2 s you traced.

How many 2 s are in the **prime factorization** of 32? _____

When a **prime factorization** has **factors** that repeat, you can use **exponents** to write the **factorization** a different way. Read the definition for **exponent.**

exponent

An **exponent** is a small, raised number written to the upper right of another number to show how many times that number is to be used as a **factor** in a multiplication expression.

Use the marker to circle the words "**small, raised number**," "**upper right**," and "**times that number is to be used as a factor**" in the definition above.

Look again at the **prime factorization** of 32.

$$2 \times 2 \times 2 \times 2 \times 2$$

How many 2 s are in the **prime factorization** of 32? _____

$2 \times 2 \times 2 \times 2 \times 2$ can also be written as 2^5

The small, raised 5 to the upper right of the **2** is an **exponent.**

Circle the **exponent** 5 and trace the 2 s in the equation below.

$$2 \times 2 \times 2 \times 2 \times 2 = 2^5$$

Write the following expressions another way using **exponents.**

$2 \times 2 \times 2 \times 2 =$ _____

$5 \times 5 \times 5 =$ _____

$3 \times 3 \times 3 \times 3 \times 3 =$ _____

Write the expression for these numbers written with **exponents**.

$7^4 =$ _____

$2^3 =$ _____

$6^5 =$ _____

2^4 5^3 3^5 $7 \times 7 \times 7 \times 7$ $2 \times 2 \times 2$ $6 \times 6 \times 6 \times 6 \times 6$

Prime Factors and Exponents *Follow the directions.*

Check your work. Did you answer all of the questions correctly?
If not, go back to Q 26 and try again.

$$6 \times 6 \times 6 \times 6 \times 6 = _____$$

$$4 \times 4 \times 4 \times 4 = _____$$

$$9 \times 9 \times 9 = _____$$

Write the expression for these numbers written with **exponents**.

$$3^2 = _____$$

$$5^3 = _____$$

$$2^4 = _____$$

Check your work. Did you answer all of the questions correctly?
If not, go back to Q 26 and try again.

$$6 \times 6 \times 6 \times 6 \times 6 = \underline{\qquad\qquad}$$

$$6^5$$

$$4 \times 4 \times 4 \times 4 = \underline{\qquad\qquad}$$

$$4^4$$

$$9 \times 9 \times 9 = \underline{\qquad\qquad}$$

$$9^3$$

Write the expression for these numbers written with **exponents**.

$$3^2 = \underline{\qquad\qquad\qquad\qquad}$$

$$3 \times 3$$

$$5^3 = \underline{\qquad\qquad\qquad\qquad}$$

$$5 \times 5 \times 5$$

$$2^4 = \underline{\qquad\qquad\qquad\qquad}$$

$$2 \times 2 \times 2 \times 2$$

Find these equations in the puzzle below.

$8^4 = 8 \times 8 \times 8 \times 8$ $\quad$ $3 \times 3 \times 3 = 3^3$ $\quad$ $5 \times 5 = 5^2$

$2^3 = 2 \times 2 \times 2$ $\quad$ $8 \times 8 \times 8 = 8^3$ $\quad$ $4^3 = 4 \times 4 \times 4$

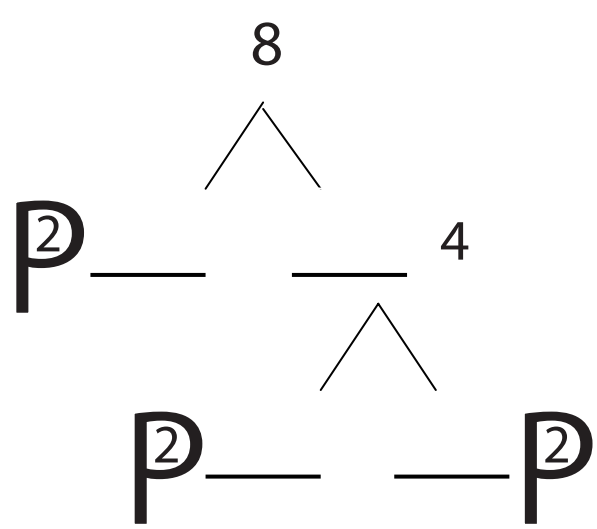
$3^2 = 3 \times 3$ $\quad$ $2 \times 2 \times 2 \times 2 = 2^4$

2	=	4^3	=	4	x	4	x	4
x	8	x	8	x	8	=	8^3	x
2	x	2	x	2	x	2	=	2^4
5	=	5	x	5	=	5^2	x	9
2^3	=	2	x	2	x	2	=	3^2
=	3^2	=	3	x	3	x	3	=
8^4	=	8	x	8	x	8	x	8
x	3	x	3	x	3	=	3^3	2

Prime Factors and Exponents *Complete the equation search.*

Now that you have learned about **factor trees**, **prime factorizations**, and **exponents**, you can combine this knowledge to write the **prime factorizations** of numbers using **exponents**.

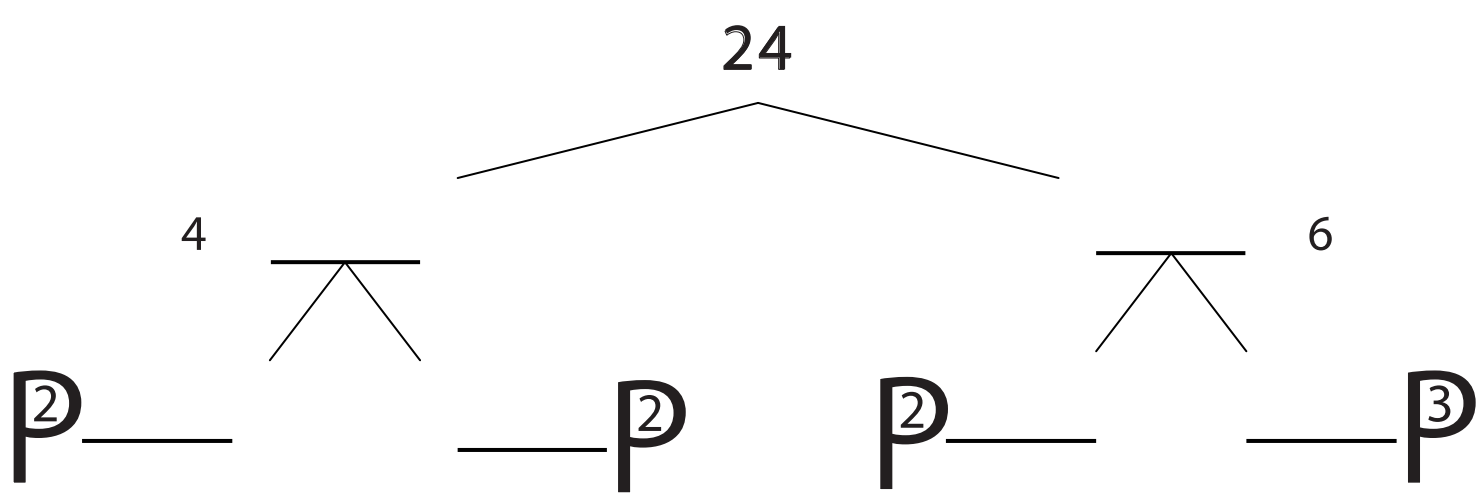
Complete the **factor tree** for 8 shown below:



Write the prime factorization of 8 two ways. _____

2 x 2 x 2 2^3

Complete the **factor trees** for 24 shown below:



Write the prime factorization of 24 two ways. _____

2 x 2 x 2 x 3 2^3x 3

Factor Trees and Exponents

Read the information and follow the instructions.

Complete a **factor tree** for each of the numbers below.
Then write the **prime factorization** of each number in TWO ways.

27

28

20

9

Factor Trees and Exponents

Follow the directions.

It's time to show how much you have learned in this Module. Use the strategies then circle the letter for the correct answer to each question.

1) Which of the following is a list containing only prime numbers?

A 27, 21, 19, 17, 15, 13

B 37, 31, 29, 23, 19, 17

C 39, 37, 31, 29, 27, 23

D 37, 33, 29, 23, 31, 19

2) Which of the following is NOT a prime number?

A 3

B 7

C 19

D 35

3) Which of the following shows 72 as the product of prime factors?

A $2^3 \times 9$

B 2×3^3

C $2^2 \times 3 \times 6$

D $2^3 \times 3^3$

Check your work. Did you answer all of the questions correctly? If not, go back to Q 30 and try again.

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